# OpenGateway: an Open-Source Gateway for Cross-Domain Agents in Complex Collaborative Tasks

**Yihao Chen**, Zian Wang, Tianshuo Hu,
Hao Dong, Junqi Li, Bingsong Wu, Yawen Wang, Xiaohui Xie, Yong Cui

Emal: {xiexiaohui, cuiyong}@tsinghua.edu.cn
GitHub: https://github.com/anAtheist987/OpenGateway

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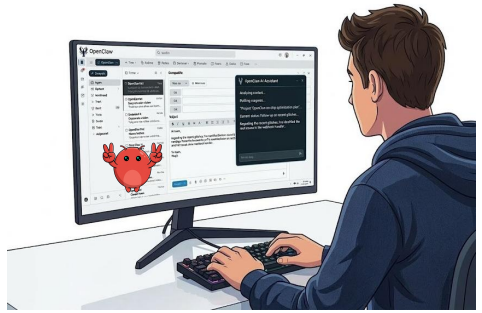
# Existing Gateway Products

## I. Focus on User-Centered AI Access

OPENCLAW

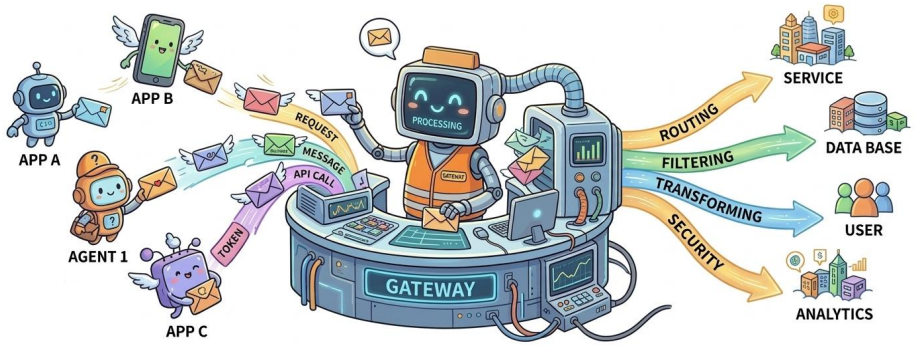
**Pro:** strong channel routing and session isolation
**Con:** external exposure and weak authentication



OpenRouter

**Pro:** flexible multi-model access
**Con:** limited support for A2A coordination

## II. Focus on General Cloud Environment




Higress

**Pro:** unified governance for agent traffic, tools,
**Con:** higher adoption and operational complexity

SOLO.IO

**Pro:** MCP support—security, observability, and control
**Con:** more control-oriented than collaboration-oriented

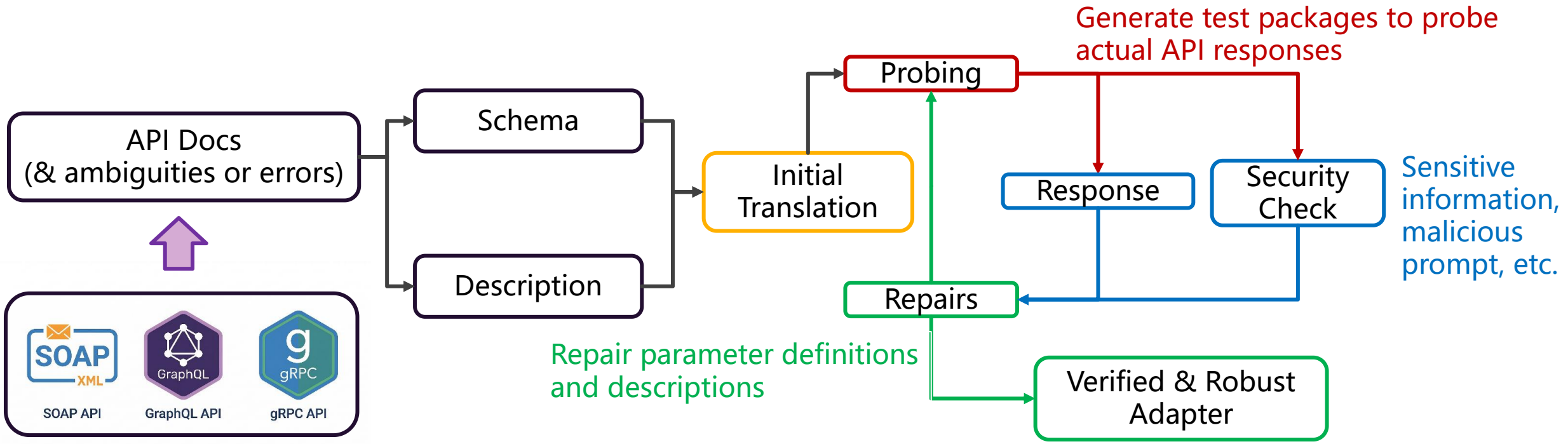
# Heterogeneous Connectivity

**Problem:**

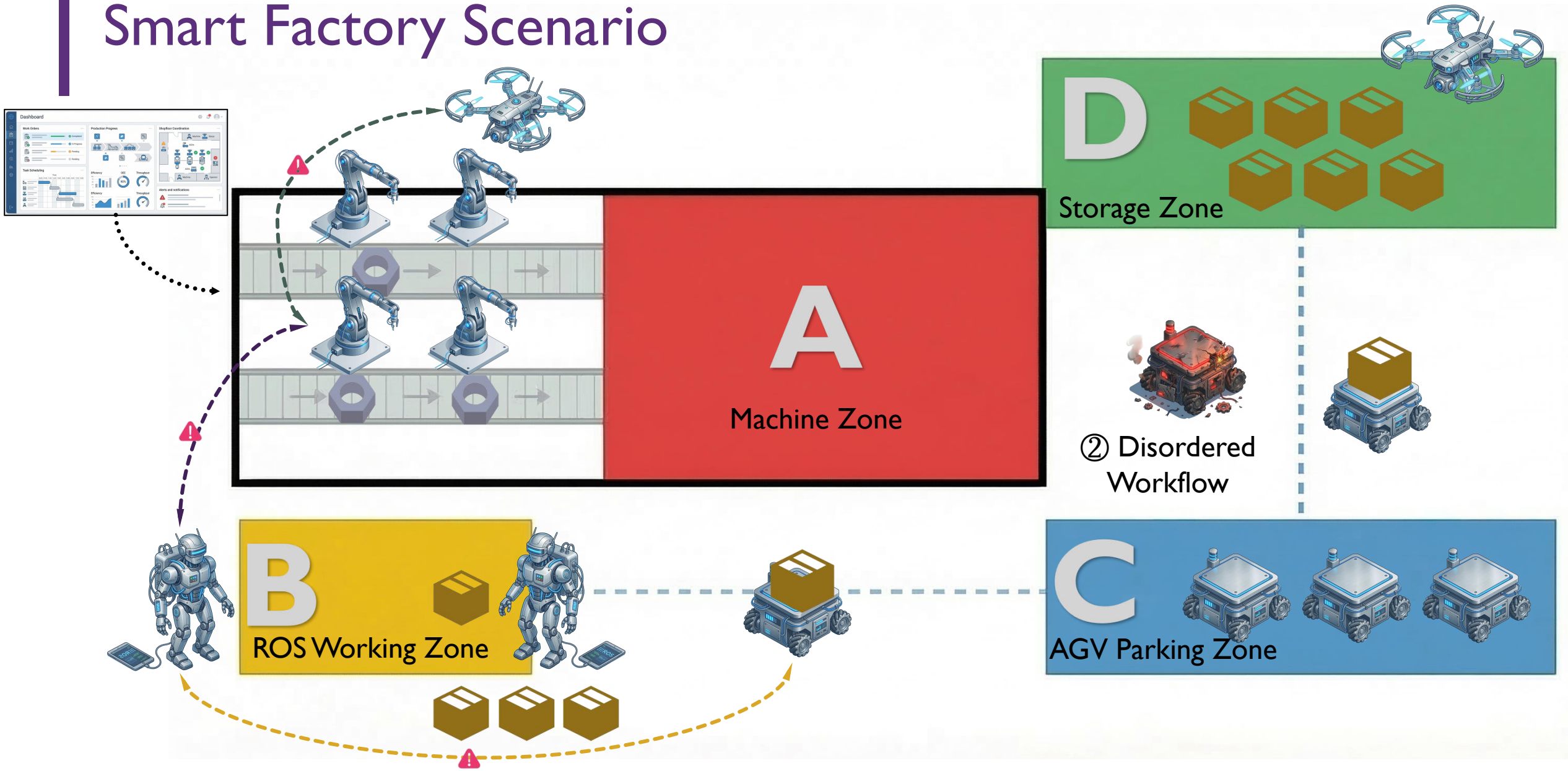
Inconsistent protocols and interfaces hinder connectivity among heterogeneous tools and agents.

**Our Approach:**

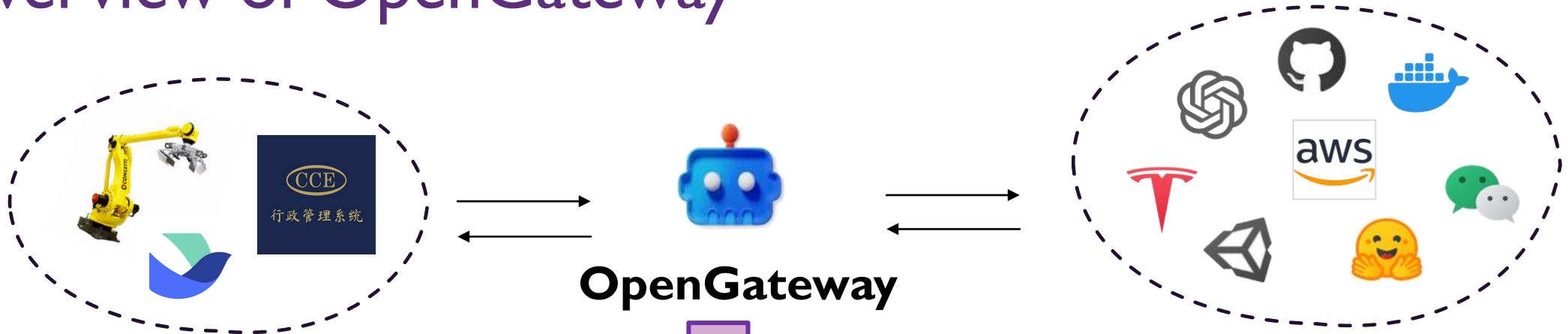
1. Capability-based registration: endpoint discovery and capability indexing
2. Self-correcting MCP translation: LLM-driven repair of schemas and mappings



# Smart Factory Scenario



# Overview of OpenGateway




**OpenGateway**

**In-domain: within the enterprise**

**Out-of-domain: public service interfaces**

**Key Functions**

**Task Orchestration**

Complex task decomposition and dynamic routing

**Working Memory**

Historical state reuse and states maintain

**Heterogeneous Connectivity**

Automatic protocol translation and agent capability discovery

**Security Assurance**

Sensitive data isolation and agent authentication

**Solve: Handling Complex Tasks**

**Solve: Connecting Heterogeneous Agents**

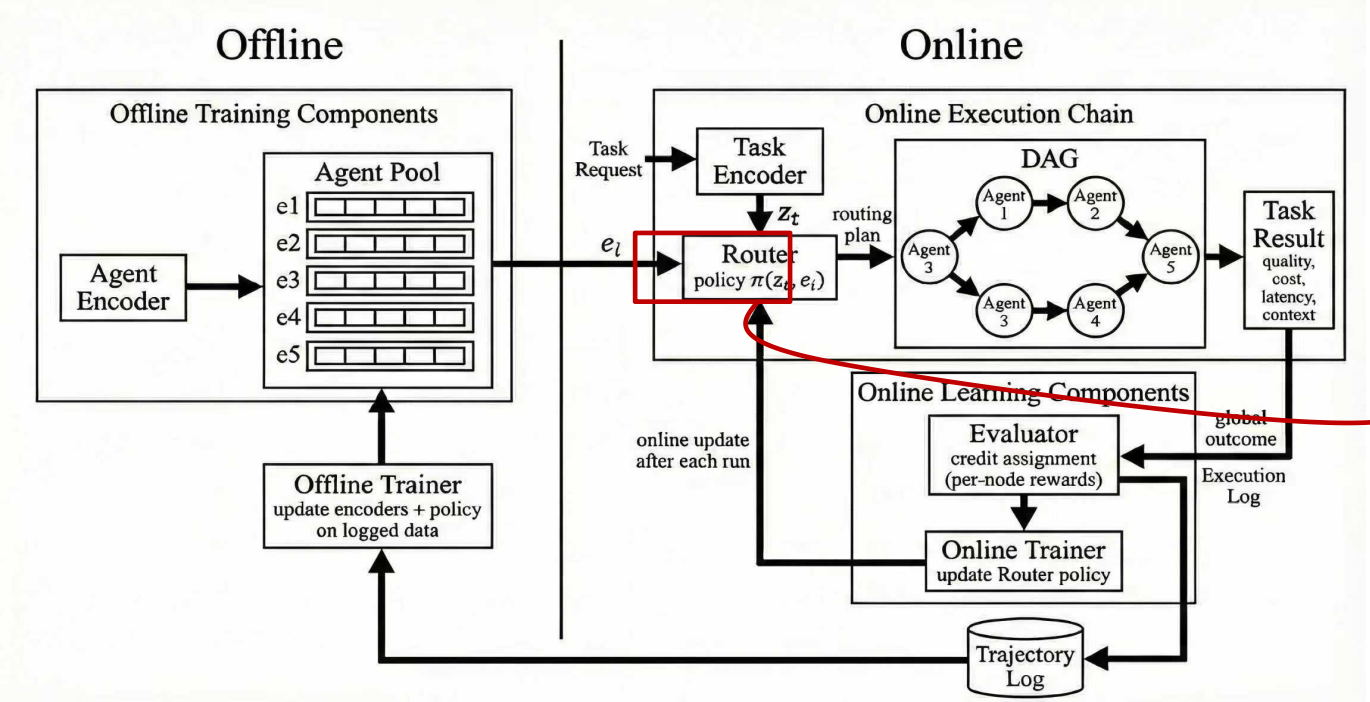
# Task Orchestration

**Problem:**

Dependencies, tool constraints, and intermediate uncertainty make one-shot generation unstable in long-horizon tasks.

**Our Approach:**

1. Replanning with intermediate feedback
2. Routing by vectorized agent profile

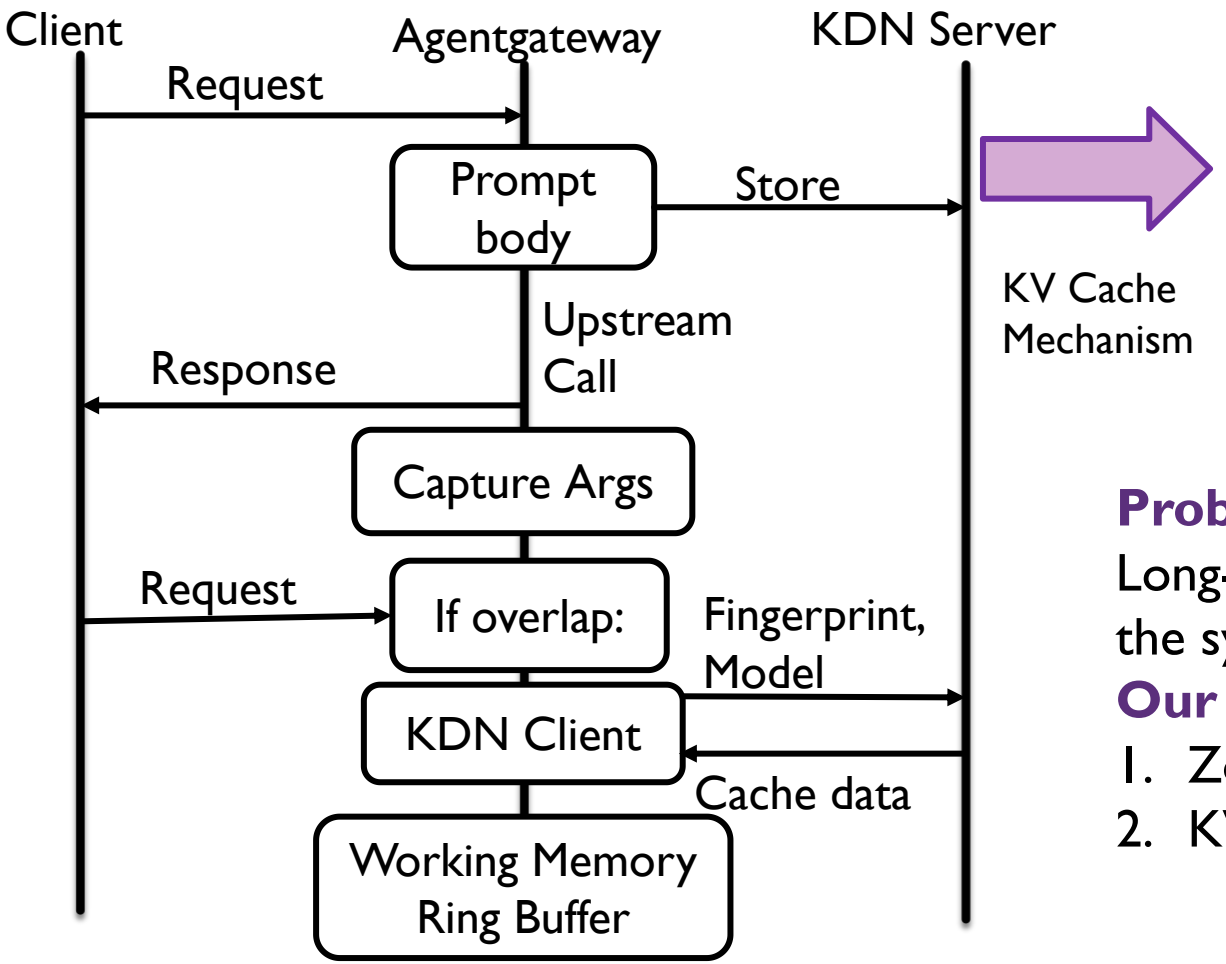



$$\pi = W \cdot \cos(z_t, e_i) + b$$

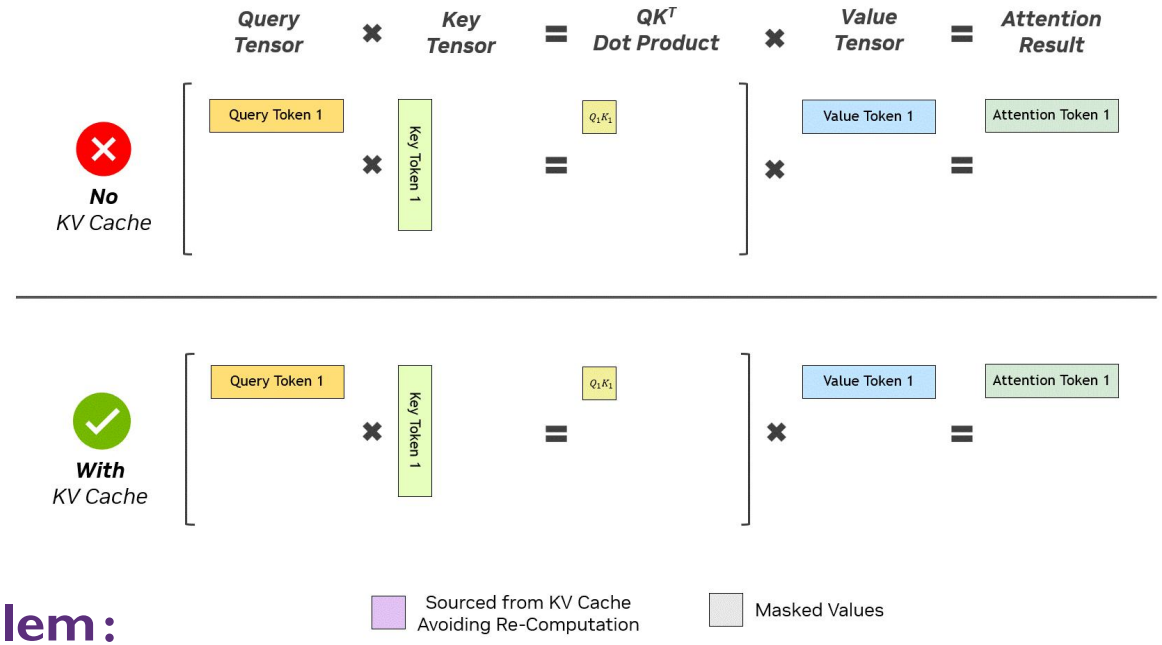
$$= W \cdot \frac{z_t^\top e_i}{\|z_t\| \|e_i\|} + b$$

$$= W \cdot \frac{\sum_{k=1}^{n} z_{t,k} e_{i,k}}{\sqrt{\sum_{k=1}^{n} z_{t,k}^2} \sqrt{\sum_{k=1}^{n} e_{i,k}^2}} + b$$

# Working Memory




KV Cache Mechanism




**Problem:**

Long-horizon execution creates more 'mental' clutter than the system can clean, leading to circular logic and lost focus.

**Our Approach:**

1. Zoned memory manages dynamic state updates
2. KV Cache reuse saves redundant computation

# Security Assurance

**Problem:**

Risks emerge from untrusted inputs, over-permissioned actions, memory contamination, and multi-step execution trajectories.

**Our Approach:**

1. Pre-execution capability assessment *(with heartbeat monitoring during execution)*
2. Sensitive information grading and rewriting

**Challenges**

- Black-box APIs
- Context Sensitivity
- Non-determinism
- Lack of Oracle
- Scalability Bottleneck

**1. Capability Modeling**
AgentCard / Skill Description
Capability Boundary Extraction

**2. Automated Test Generation**
Task Synthesis
Metamorphic Prompts

**3. Black-box Interaction & Logging**
Challenge Response
Response Logging

**4. Scoring & Verification**
Metamorphic Relations
CARE Scoring

# Introduction Of OpenGateway

Open-source project address: https://github.com/anAtheist987/OpenGateway

- **Background**
- Connected Objects
- User Interaction
- Task Orchestration
- Working Memory
- Conclusion

# Our Potential Clients

**Scenario:**

Enterprise management of internal agents

**Needs:**

A middleware layer that manages agents and enables collaboration between them

# Deployment Features & Value

## Deployment Mode

deployed on edge servers and oriented to cross-domain ToB service

## Data Privacy

support personalized information security levels

## Computing Demand

at least 9B-parameter large language model

## Token Savings

approx. 30%–60% LLM token savings

## Model Selection

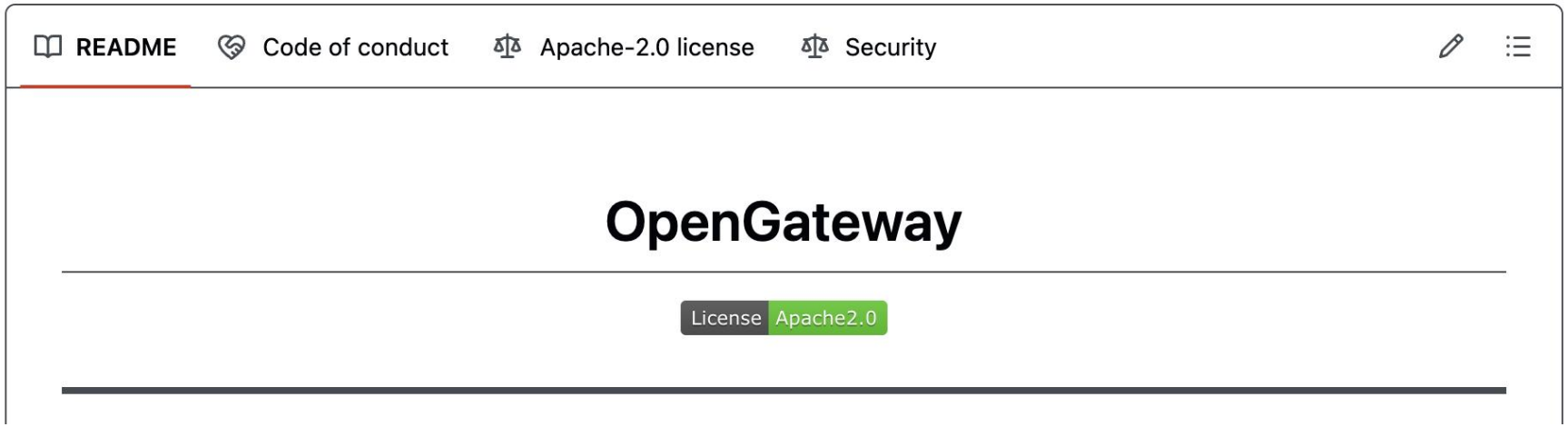
support local model deployment & external API access

## Complex Tasks

identify task difficulty and automate decomposition

# Open Source Code & Community

- **Ready to Use :** Task Orchestration & Working Memory
- **Coming Soon :** Agent Registration, Automatic Cross-Protocol Translation, etc.
- **Join Us:** Try https://github.com/anAtheist987/OpenGateway now !
- **Active IETF Draft :** https://www.ietf.org/archive/id/draft-agent-gw-01.html



We welcome your feedback and comment!
We are also looking for collaboration!
Thank You!

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